WALL JACK

Connection to the telephone network

PUBLIC SWITCHED TELEPHONE NETWORK

Loudspeaker reproduces transmitted speech in telephone earpiece

CALLER

**4 Sending the sound signal**
The signals travel rapidly through a temporary connection formed across a worldwide telephone communications network known as the public switched telephone network (PSTN). Signals may travel via fiber-optic cables, wires, satellite dishes, and cell towers between the caller's and receiver's telephones.

**3 Making a sound signal**
Once the telephones are connected, the caller speaks into a microphone in the mouthpiece, thus producing sound waves. The waves cause a membrane to vibrate and create electrical signals, which travel down the line.

MOUTHPIECE

Microphone transforms sound waves into electrical signals

RECEIVER

EARPIECE

**5 Reproducing the sound**
Inside the earpiece is a speaker. When it receives electrical signals, a membrane is driven back and forth at frequencies matching the current, causing the air to vibrate and produce sound waves.

**"AHOY" WAS BELL'S SUGGESTED GREETING ON THE TELEPHONE BUT WAS REPLACED BY THOMAS EDISON'S SUGGESTION "HELLO"**

ELECTRIC CABLE

BOOSTER AMPLIFIER

Consists of transistors, which increase the power of the signal, extending its range

**2 Electric cable**
Electrical signals from the microphone are amplified and sent through electric cables. This is a slower method of transmission than radio waves.

Modulated radio signal radiates through the air and is detected by aerial

AERIAL

RADIO TRANSMISSION

**2 Radio transmission**
The signal is modulated using a radio-frequency carrier wave (see pp.180–181), which is produced by an oscillator. The signal is then transmitted wirelessly from an antenna in the form of radio waves.

FIBER-OPTIC CABLE

Core made of plastic or glass

Light beam bounces off walls

Plastic outer coating

Cladding keeps light signal inside core

**2 Optical fibers**
The signal is combined with light produced by a laser beam, which travels through fiber-optic cables.

AMPLIFIER

RADIO DETECTOR

AMPLIFIER

LIGHT DETECTOR

**3 Sound signal arrives**
The signals arrive at their destination and are passed on to a telephone receiver. The receiver demodulates the signal, extracting from it the useful information and reproducing the sound.